

CS Revise — Stage 2, Track E: Auto-Marking

Submission Summary

What I did

I researched three approaches to auto-marking short written answers, built a working prototype that marks a student answer against a mark scheme, tested it against 14 hand-written sample answers across two GCSE/11+ style questions, and wrote up a design plan for how this would plug into CS Revise's real marking flow, along with an honest risk assessment.

Approach

After comparing keyword matching, embedding/similarity matching, and LLM-based grading (full comparison in `research_comparison.md`), I built a hybrid of the first two: the prototype checks each mark-scheme point using both a small keyword/synonym list and a TF-IDF text-similarity score, and awards a mark if either signal is confident enough. This choice was deliberate: it runs instantly, needs no API key or internet connection, and — unlike a pure LLM approach — every decision can be explained to a teacher in plain terms ("this point was awarded because the answer matched keyword X" / "...because it was semantically similar to the exemplar phrase").

The design note proposes using this fast, free engine for the clear-cut majority of answers, and escalating only the borderline cases the engine itself flags to a stronger (LLM-based) check — rather than either avoiding LLMs entirely or calling one for every single answer.

Results (tested honestly, not cherry-picked)

1. 10/14 (71%) exact mark match across the full test set, 100% within 1 mark
2. 100% exact match on full-mark answers; the weakest category was partial-mark answers (40%)
3. Main failure mode identified: the engine over-credits answers that mention the right topic words without making the actual causal explanation a mark point wants — a limitation reported in full with examples, in `accuracy_report.md`

Key assumptions made

- Mark schemes can be broken into discrete points, each with keywords and a short exemplar phrase — this matches how many GCSE mark schemes are already structured
- A prototype should be judged on whether it's honest about its limitations, not just its headline accuracy number — so the test set deliberately includes answers designed to expose weaknesses, not just easy cases
- Given the 72-hour window, a fully offline, zero-cost approach was the right thing to build and test properly, with the LLM-based approach specified as a designed (not built) escalation path for the cases the cheaper engine is genuinely unsure about
- Human review remains mandatory for flagged/borderline cases in any real deployment — this prototype is designed as a first-pass assistant to a teacher, not a replacement for one